

An Analysis of Chain of thought reasoning

Chain of thought reasoning

Chain of thought reasoning -- Part 1

History Early precedents of structured user interaction with ruled-based AI systems can be found in enterprise automation software from 1990s. For example, The Intelligent Filling Manager (1999), developed by Krishna C. Mukherjee, used a dynamic Q&A interface driven by rule-based expert system to collect user inputs for generating regulatory filings automatically across jurisdictions. While not involving neural networks, such systems featured prompt-like workflows that influenced later human-in-the-loop AI designs. In 2018, researchers first proposed that all previously separate tasks in natural language processing (NLP) could be cast as question-answer problems over a context. In addition, they trained a first single, joint, multi-task model that would answer any task-related question like "What is the sentiment" or "Translate this sentence to German" or "Who is the president?" The AI boom saw an increased focus within academic literature and professional practice on applying prompting techniques to get the model to output the desired outcome and avoid nonsensical output, a process characterized by trial-and-error. After the release of ChatGPT in 2022, prompt engineering was soon seen as an important business skill; companies began hiring dedicated prompt engineers, although, given advances in AI's ability to generate prompts better than humans, the employment market for prompt engineers has faced uncertainty. According to The Wall Street Journal in 2025, the job of prompt engineer was one of the hottest in 2023, but has become obsolete due to models that better intuit user intent and to company trainings. A repository for prompts reported that over 2,000 public prompts for around 170 datasets were available in February 2022. In 2022, the chain-of-thought prompting technique was proposed by Google researchers. In 2023, several text-to-text and text-to-image prompt databases were made publicly available. The Personalized Image-Prompt (PIP) dataset, a generated image-text dataset that has been categorized by 3,115 users, has also been made available publicly in 2024.

Chain of thought reasoning -- Part 2

Textual inversion and embeddings For text-to-image models, textual inversion performs an optimization process to create a new word embedding based on a set of example images. This embedding vector acts as a "pseudo-word" which can be included in a prompt to express the content or style of the examples.

Chain of thought reasoning -- Part 3

Image prompting In 2023, Meta's AI research released Segment Anything, a computer vision model that can perform image segmentation by prompting. As an alternative to text prompts,

Segment Anything can accept bounding boxes, segmentation masks, and foreground/background points.

Chain of thought reasoning -- Part 4

Chain-of-thought (CoT) prompting is a technique that allows large language models (LLMs) to solve a problem as a series of intermediate steps before giving a final answer. In 2022, Google Brain reported that chain-of-thought prompting improves reasoning ability by inducing the model to answer a multi-step problem with steps of reasoning that mimic a train of thought. Chain-of-thought techniques were developed to help LLMs handle multi-step reasoning tasks, such as arithmetic or commonsense reasoning questions. When applied to PaLM, a 540 billion parameter language model, according to Google, CoT prompting significantly aided the model, allowing it to perform comparably with task-specific fine-tuned models on several tasks, achieving state-of-the-art results at the time on the GSM8K mathematical reasoning benchmark. It is possible to fine-tune models on CoT reasoning datasets to enhance this capability further and stimulate better interpretability. As originally proposed by Google, each CoT prompt is accompanied by a set of input/output examples—called exemplars—to demonstrate the desired model output, making it a few-shot prompting technique. However, according to a later paper from researchers at Google and the University of Tokyo, simply appending the words "Let's think step-by-step" was also effective, which allowed for CoT to be employed as a zero-shot technique.

Chain of thought reasoning -- Part 5

Prompt engineering is the process of structuring natural language inputs (known as prompts) to produce specified outputs from a generative artificial intelligence (GenAI) model. Context engineering is the related area of software engineering that focuses on the management of non-prompt contexts supplied to the GenAI model, such as metadata, API tools, and tokens. It can also be defined as the practice of designing and refining input instructions given to a generative AI model to produce more accurate, relevant, or useful outputs. Effective prompt engineering involves understanding how a model interprets language, and may include techniques such as few-shot prompting, chain-of-thought prompting, and role assignment. It is increasingly considered a skill for working with large language models (LLMs) in both research and professional contexts. During the 2020s AI boom, prompt engineering became regarded as a business capability across corporations and industries. Employees with the title prompt engineer were hired to create prompts that would increase productivity and efficacy, although the individual title has since lost traction amid AI models that

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produce better prompts than humans and corporate training in prompting for general employees. Common prompting techniques include multi-shot, chain-of-thought, and tree-of-thought prompting, as well as the use of assigning roles to the model. Automated prompt generation methods, such as retrieval-augmented generation (RAG), provide for greater accuracy and a wider scope of functions for prompt engineers. Prompt injection is a type of cybersecurity attack that targets machine learning models through malicious prompts.

Chain of thought reasoning -- Part 6

GraphRAG (coined by Microsoft Research) is a technique that extends RAG with the use of a knowledge graph to allow the model to connect disparate pieces of information, synthesize insights, and understand summarized semantic concepts over large data collections. It was shown to be effective on datasets like the Violent Incident Information from News Articles.

Chain of thought reasoning -- Part 7

Prompt injection is a cybersecurity exploit in which adversaries craft inputs that appear legitimate but are designed to cause unintended behavior in machine learning models, particularly large language models. This attack takes advantage of the model's inability to distinguish between developer-defined prompts and user inputs, allowing adversaries to bypass safeguards and influence model behaviour. While LLMs are designed to follow trusted instructions, they can be manipulated into carrying out unintended responses through carefully crafted inputs.

Chain of thought reasoning -- Part 8

Techniques Common terms used to describe various specific prompt engineering techniques include chain-of-thought, tree-of-thought, and retrieval-augmented generation (RAG). A 2024 survey of the field identified over 50 distinct text-based prompting techniques, 40 multimodal variants, and a vocabulary of 33 terms used across prompting research, highlighting a present lack of standardised terminology for prompt engineering. Vibe coding is an AI-assisted software development method where a user prompts an LLM with a description of what they want and lets it generate or edit the code. In 2025, "vibe coding" was the Collins Dictionary word of the year.